



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 9	
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Batch	2023 - 2027
Class	CSE – B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Container Management & Networking
Theory	Docker container management is the process of creating, running, monitoring, stopping, inspecting, and removing containers. Port mapping allows applications running inside a container to be accessed from the host machine by mapping a host port to a container port. Docker provides commands such as <code>docker ps</code> to view running containers, <code>docker inspect</code> to examine detailed container configuration and network settings, <code>docker stop</code> to gracefully stop a container, and <code>docker rm</code> to remove unused containers. These commands help efficiently manage the lifecycle of Docker containers and verify their network configuration and status. Commands Used - Created and started the Docker container in detached mode with port mapping: <code>docker run -d -p 8081:5000 --name flask-network-demo my-flask-app</code> - Listed all running Docker containers: <code>docker ps</code> - Stopped the running Docker container: <code>docker stop flask-network-demo</code> - Listed all Docker containers, including stopped containers: <code>docker ps -a</code> - Inspected the container's network configuration and settings: <code>docker inspect flask-network-demo</code> - Removed the stopped Docker container: <code>docker rm flask-network-demo</code> Verified that the container had been removed: <code>docker ps -a</code>
Implementation	Step 1: Run a Container with Port Mapping

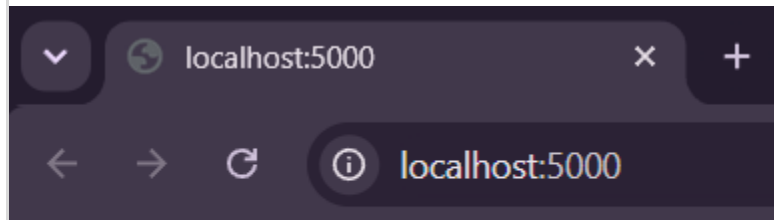
```

PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app> docker run -p 5000:5000 my-flask-app
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.3:5000
Press CTRL+C to quit

```

Started Docker Container — Started the Docker container named flask-network-demo in detached mode and mapped host port 8081 to container port 5000, allowing the application to run in the background.

Step 2: Access the Application via Your Browser



Verified Application Access — Accessed the Dockerized Flask application through `http://localhost:8081` and verified that the application was successfully running inside the Docker container.

Step 3: List Running Containers

```

PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
177e15418f71   docker-n8n.io/n8nio/n8n  "tini -- /docker-ent..."  2 weeks ago   Up About a minute   0.0.0.0:5678->5678/tcp, [::]:5678->5678/tcp   n8n
PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app>

```

Listed Running Containers — Displayed all currently running Docker containers using the `docker ps` command and verified that the flask-network-demo container was running with port mapping from 8081 on the host to 5000 inside the container.

Step 4: Stop the Running Container

```

PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app> docker stop
docker: 'docker stop' requires at least 1 argument

Usage: docker stop [OPTIONS] CONTAINER [CONTAINER...]

See 'docker stop --help' for more information
PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app>

```

Stopped Docker Container — Stopped the running flask-network-demo container using the `docker stop` command, terminating the application gracefully.

Step 5: Inspect the Container's Network Settings

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app> docker inspect 177e15418f71

[{"IPAddress": "172.17.0.2",
  "MacAddress": "ae:5b:60:7a:9b:09",
  "IPPrefixLen": 16,
  "IPv6Gateway": "",
  "GlobalIPv6Address": "",
  "GlobalIPv6PrefixLen": 0,
  "DNSNames": null
}]
},
"ImageManifestDescriptor": {
  "mediaType": "application/vnd.oci.image.manifest.v1+json",
  "digest": "sha256:60dac72c8a23a3ad0921fb9e8e4cd8a67981b4377ea2caca7c672bed6c0c6886",
  "size": 2560,
  "platform": {
    "architecture": "amd64",
    "os": "linux"
  }
}
}
]
PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app> |
```

Inspected Container Network Settings — Used the docker inspect command to view detailed information about the flask-network-demo container, including its NetworkSettings and network configuration.

Step 6: Remove the Stopped Container

```
PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app> docker rm flask-network-demo
Error response from daemon: No such container: flask-network-demo
PS C:\Users\sanyu\OneDrive\Desktop\Studies\DevOps\my-flask-app> docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
35fb88b21333	my-flask-app	"python app.py"	5 minutes ago	Exited (0) 3 minutes ago	
177e15418f71	docker.io/n8nio/n8n	"tini -- /docker-ent..."	2 weeks ago	Up 5 minutes	0.0.0.0:5678->5678/tcp, [::]:5678->5678/tcp
4aa8b212ae85	n8nio/n8n:latest	"tini -- /docker-ent..."	2 weeks ago	Exited (0) 2 weeks ago	
10418295ec7	n8nio/n8n:latest	"tini -- /docker-ent..."	2 weeks ago	Exited (255) 7 days ago	5678/tcp
80003839101	n8nio/n8n:latest	"tini -- /docker-ent..."	2 weeks ago	Exited (0) 2 weeks ago	
6fbfc732bf3a	my-flask-app	"python app.py"	3 weeks ago	Created	
05ae9e2e78d9	my-flask-app	"python app.py"	3 weeks ago	Exited (0) 3 weeks ago	
4e589a12ed78	8782003ec4c6	"python app.py"	3 weeks ago	Exited (0) 3 weeks ago	
97f70822ddc4	deb6949154db	"python app.py"	3 weeks ago	Exited (0) 3 weeks ago	
235ad584e008	8782003ec4c6	"python app.py"	3 weeks ago	Exited (0) 3 weeks ago	
bdf71bf32e	8782003ec4c6	"python app.py"	3 weeks ago	Exited (0) 3 weeks ago	
339f3839f93a	deb6949154db	"python app.py"	3 weeks ago	Exited (0) 3 weeks ago	
e999b23981ea	8c1481b8bec2	"python app.py"	3 weeks ago	Exited (0) 3 weeks ago	
2605d6785426	8c1481b8bec2	"python app.py"	3 weeks ago	Exited (0) 3 weeks ago	

Removed Docker Container — Removed the flask-network-demo container using the docker rm command and verified its successful removal by confirming that it no longer appeared in the list of Docker containers.

Subject Faculty Name: Prof. Arijit Dutta